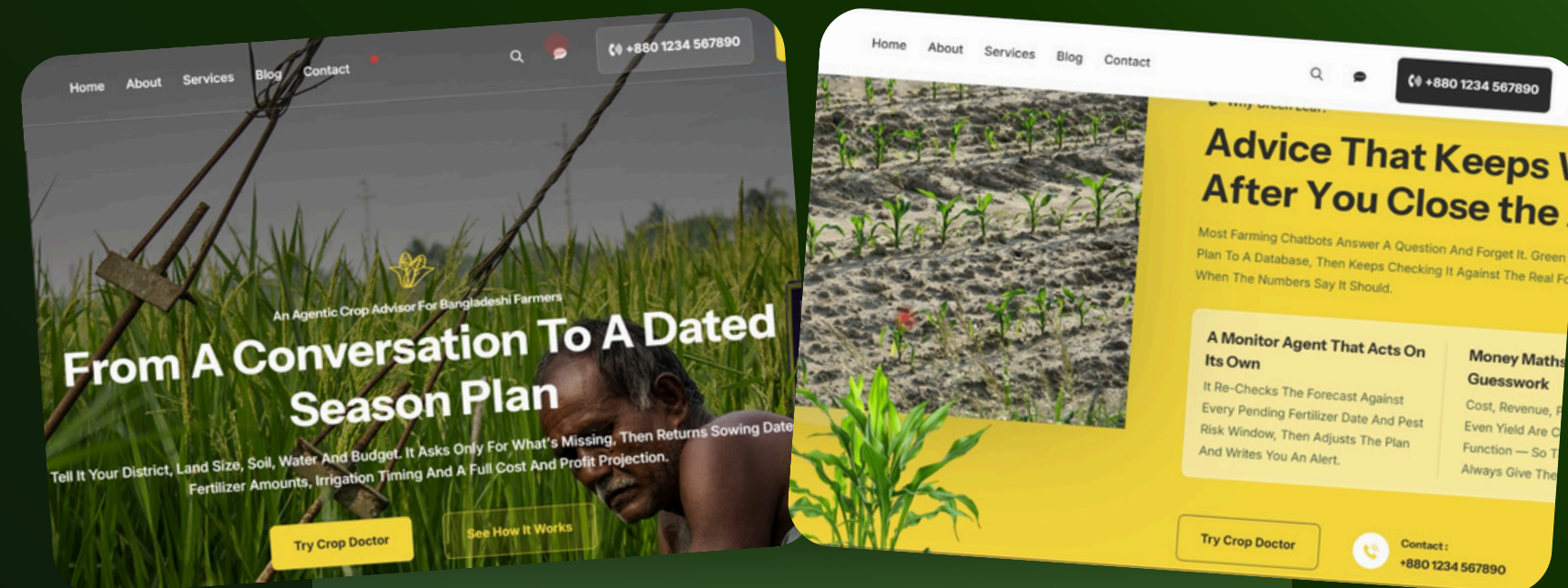




Agrotech Revolution

An multi-agent ai
powered agrotech
solution



Presented by

TEAM ORBIT

A New Agrotech era



an AI-powered multi-agent agricultural advisor for Bangladeshi farmers. It uses natural conversation, live weather data, and trusted agricultural references to recommend crops, create personalized season plans, estimate costs and profits, and automatically adjust plans when weather risks arise—all in Bangla or English.

The Rise of Greenleaf

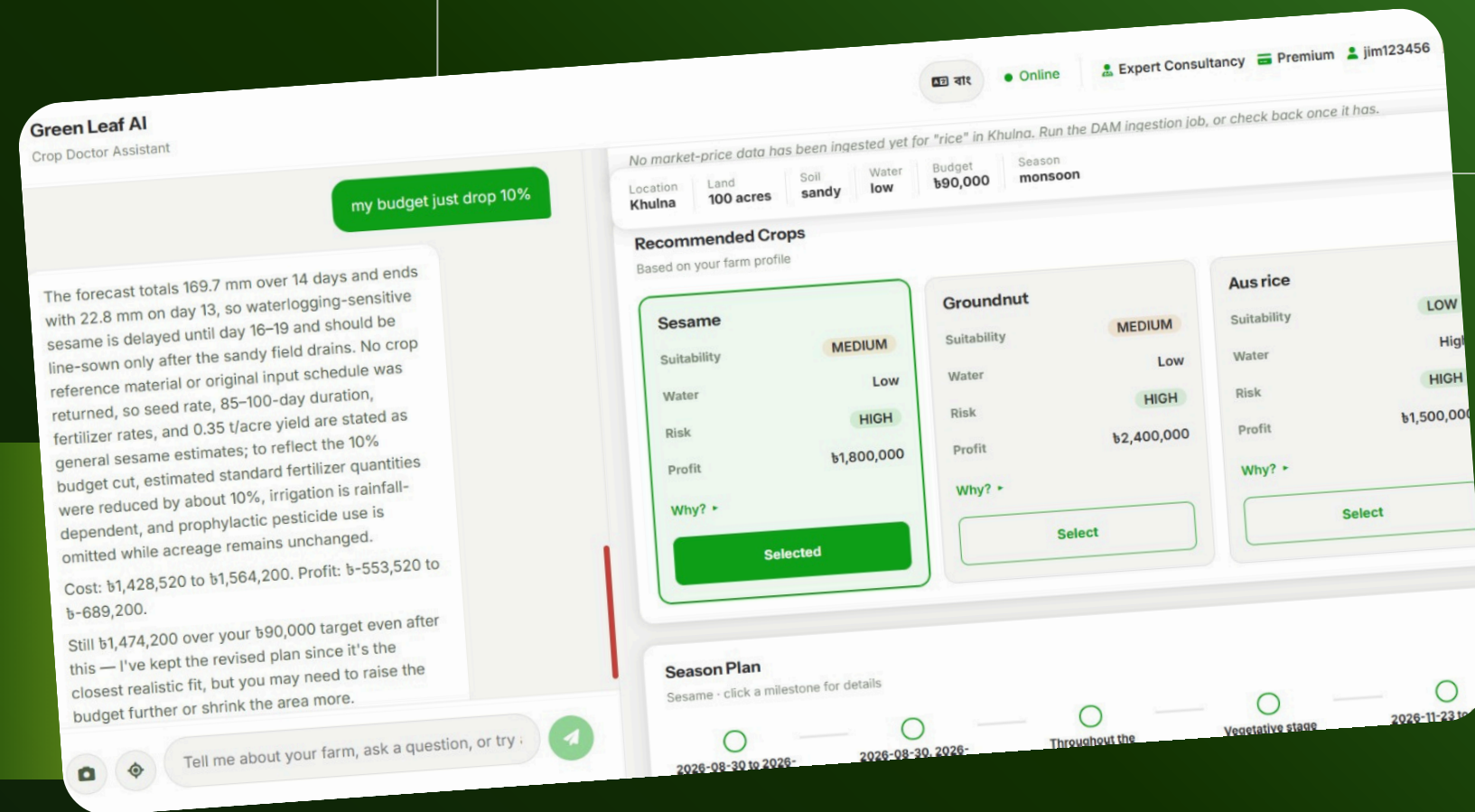
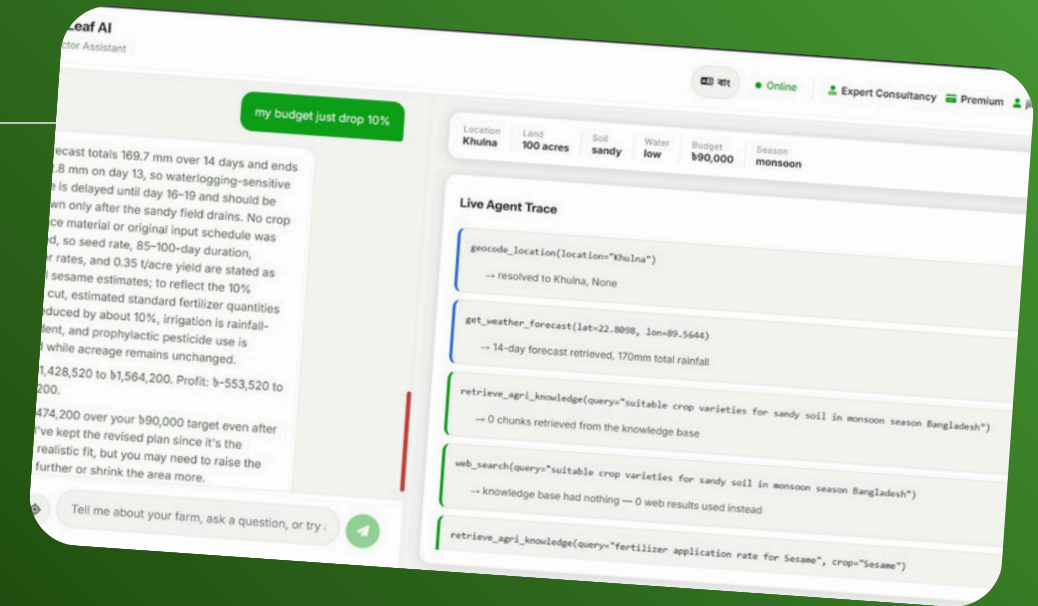
KEY HIGHLIGHTS

- Monitoring agent
- Personalized crop prediction
- Multilingual farmer support
- Season plan with profit estimates

Greenleaf in Action

Everyday Applications

- Langraph based conversation & monitoring agent
- Natural language classification
- Rag based agronomy document embedding
- live weather retrival through open-meteo
- Deterministic cost profit ROI & Break-even analysis



Greenleaf in Action

Everyday Applications

Multilingual farmer support

Recording... tap the mic again to stop and send



Tell me about your farm, ask a question, or try :



Voice message

আমি কৃষিকাজ শুরু করতে চাই, আমাকে একটা প্ল্যান দাও.



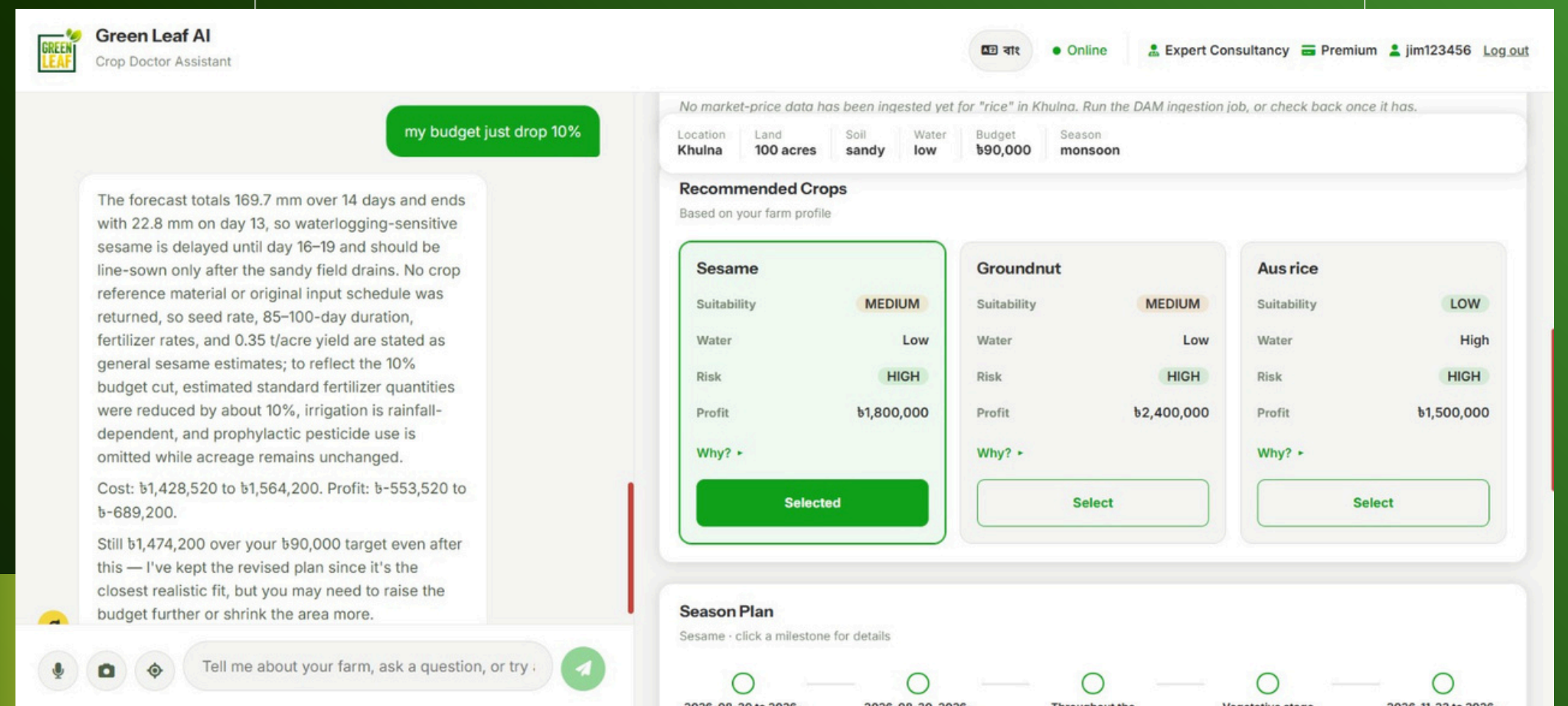
Thinking...



Tell me about your farm, ask a question, or try :



Everyday Applications



Greenleaf in Action

Everyday Applications

Agent in action based on
user situation

 **Green Leaf AI**
Crop Doctor Assistant

৳ বাট

Online

Expert Consultancy

Premium

jim123456

Log out

my budget just drop 10%

The forecast totals 169.7 mm over 14 days and ends with 22.8 mm on day 13, so waterlogging-sensitive sesame is delayed until day 16-19 and should be line-sown only after the sandy field drains. No crop reference material or original input schedule was returned, so seed rate, 85-100-day duration, fertilizer rates, and 0.35 t/acre yield are stated as general sesame estimates; to reflect the 10% budget cut, estimated standard fertilizer quantities were reduced by about 10%, irrigation is rainfall-dependent, and prophylactic pesticide use is omitted while acreage remains unchanged.

Cost: ₳1,428,520 to ₳1,564,200. Profit: ₳-553,520 to ₳-689,200.

Still ₳1,474,200 over your ₳90,000 target even after this — I've kept the revised plan since it's the closest realistic fit, but you may need to raise the budget further or shrink the area more.

Location	Land	Soil	Water	Budget	Season
Khulna	100 acres	sandy	low	₳90,000	monsoon

Live Agent Trace 18 calls

geocode_location(location="Khulna")
→ resolved to Khulna, None

get_weather_forecast(lat=22.8098, lon=89.5644)
→ 14-day forecast retrieved, 170mm total rainfall

retrieve_agri_knowledge(query="suitable crop varieties for sandy soil in monsoon season Bangladesh")
→ 0 chunks retrieved from the knowledge base

web_search(query="suitable crop varieties for sandy soil in monsoon season Bangladesh")
→ knowledge base had nothing — 0 web results used instead

retrieve_agri_knowledge(query="fertilizer application rate for Sesame", crop="Sesame")

   Tell me about your farm, ask a question, or try : 



Greenleaf in Action

Everyday Applications

Dynamic financial analysis
updates when cost
changes

Location	Land	Soil	Water	Budget	Season
Rajshahi	1.2 acres	loamy	medium	₹30,000	winter
Labor					₹9,000 ₹5,400
Post-harvest & Transport					₹3,000 ₹1,800
Total Cost					
₹24,157					
₹40,482 ↘ -₹16,325					
Revenue					
₹63,744					
₹106,240 ↘ -₹42,496					
Net Profit					
₹39,587					
₹65,758 ↘ -₹26,171					
ROI					
164%					
162% ↗ +2.0%					
Break-even					
0.75 tons					
1.27 tons ↘ -0.52					

Ask in the chat — e.g. "what if my budget drops 40%?" — and the agent will recalculate this against your actual plan.



TechStack in Action

Everyday Applications

PostgreSQL

React 19

FastAPI

Tavily API

OpenAI API

Open-Meteo
API

Kindwise
crop.health API

LangGraph

Docker

JWT

Whisper

DAM API





Thank You

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